

[Lab for Microsoft Windows operating system]

Installation of Python, and virtualenv

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1 Python installation

- Download the right version of Python interpreter from <https://www.python.org/downloads/windows/> for Microsoft Windows considering its number of bits (e.g., 32 bit or 64 bit).
- Assign the location of the downloaded Python and the corresponding Pip to your system's environment variable PATH (for both *user variable* and *system variable*). Make these two locations be on the highest level in the PATH lists.
- Verify that Python version that you have just installed is correctly called, by opening a terminal and typing:

```
python --version
```

2 Virtualenv installation

- Open a terminal (e.g., PowerShell with administrator right)
- Install *virtualenv* for Python virtualization:

```
python -m pip install virtualenv
```

- Make sure that the execution of PowerShell scripts is enabled by typing in a terminal

```
Get-ExecutionPolicy
```

See that the returned value is *Unrestricted*. Then you are good to continue. However, if the answer was *Restricted*, then type the following command in a terminal:

```
Set-ExecutionPolicy unrestricted
```

Then, accept the change to be made.

- Now, go to your Python working directory and type in a terminal: *python -m virtualenv -python python39 venv39*

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```
python -m virtualenv my_venv
```

- Then, activate the virtual environment that you have just created (i.e., *my_venv*) by typing in a terminal:

```
.\my_venv\Scripts\activate
```

- Now, your Python virtual environment should be activated. You are now ready to install any Python libraries or code own Python scripts for your project with the Python version that you selected.
- To quit the virtual environment, simply type in a terminal:

```
deactivate
```